

C++ Programming – Comprehensive Syllabus

Course Overview

This course provides a comprehensive introduction to C++ programming, covering procedural programming, object-oriented programming (OOP), Standard Template Library (STL), file handling, exception handling, and basic data structures. Students will develop problem-solving skills and build real-world applications using modern C++.

Duration: 8–12 Weeks (50–70 Hours)

Prerequisites:

- Basic computer knowledge
 - No prior programming experience required (basic C knowledge is recommended)
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Module 1: Introduction to C++ Programming (Week 1)

Topics

- Introduction to C++
- Features of C++
- Installing Compiler and IDE
- Structure of a C++ Program
- Variables and Data Types
- Constants
- Input and Output (cin, cout)
- Operators
- Comments and Naming Conventions

Practical Exercises

- Hello World Program
 - Basic Calculator
 - Unit Conversion Programs
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Module 2: Decision Making & Looping (Week 2)

Topics

- if, if-else, Nested if
- switch Statement
- for Loop
- while Loop
- do-while Loop
- break and continue
- Scope of Variables

Practical Exercises

- Grade Calculator
 - Leap Year Checker
 - Number Guessing Game
 - Multiplication Table Generator
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Module 3: Functions & Recursion (Week 3)

Topics

- Function Declaration and Definition
- Function Overloading
- Default Arguments
- Inline Functions
- Recursive Functions
- Storage Classes
- Header Files

Practical Exercises

- Factorial Using Recursion
 - Fibonacci Series
 - Prime Number Checker
 - Menu-Driven Calculator
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Module 4: Arrays, Strings & Pointers

(Week 4)

Topics

- One-Dimensional Arrays
- Multidimensional Arrays
- Character Arrays
- C++ String Class
- Pointer Basics
- Pointer Arithmetic
- Dynamic Memory Allocation (new and delete)

Practical Exercises

- Matrix Operations
 - String Manipulation
 - Array Sorting
 - Dynamic Memory Programs
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Module 5: Object-Oriented Programming

(Week 5)

Topics

- Classes and Objects
- Constructors
- Destructors
- Member Functions
- Access Specifiers
- Static Members
- Friend Functions
- this Pointer

Practical Exercises

- Student Management System
 - Bank Account Class
 - Employee Information System
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Module 6: Advanced OOP Concepts (Week 6)

Topics

- Inheritance
- Types of Inheritance
- Polymorphism
- Function Overriding
- Virtual Functions
- Abstract Classes
- Encapsulation
- Data Abstraction

Practical Exercises

- Library Management System
 - Vehicle Management System
 - Shape Area Calculator
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Module 7: Templates, Exception Handling & File Handling (Week 7)

Topics

- Function Templates
- Class Templates
- Exception Handling (try, catch, throw)
- File Streams (ifstream, ofstream, fstream)
- Binary and Text Files

Practical Exercises

- Generic Calculator
 - File-Based Student Database
 - Exception Handling Programs
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Module 8: Standard Template Library (STL) & Data Structures (Week 8)

Topics

- Introduction to STL
- Vectors
- Lists
- Queues
- Stacks
- Maps
- Sets
- Iterators
- Algorithms (sort, find, binary_search)

Practical Exercises

- Student Record System Using STL
 - Contact Management System
 - Sorting and Searching Applications
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Module 9: Mini Project (Week 9)

Choose one:

- Library Management System
 - Banking Management System
 - Employee Payroll System
 - Hospital Management System
 - Inventory Management System
 - Student Information System
 - Hotel Reservation System
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Module 10–12: Capstone Project

Complete Software Development Project

- Requirement Analysis
- Object-Oriented Design

- UML Basics
 - Class Diagram Design
 - Application Development
 - File-Based Data Management
 - Testing and Debugging
 - Documentation
 - Final Demonstration
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Tools & Software

- GCC / G++
 - Visual Studio Code
 - Code::Blocks
 - Dev-C++
 - CLion
 - Microsoft Visual Studio
 - Git & GitHub
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Core Concepts Covered

- Programming Fundamentals
 - Object-Oriented Programming (OOP)
 - Functions and Recursion
 - Arrays and Strings
 - Pointers and Dynamic Memory
 - Classes and Objects
 - Inheritance and Polymorphism
 - Templates
 - Exception Handling
 - File Handling
 - Standard Template Library (STL)
 - Basic Data Structures and Algorithms
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Assessment Scheme

Component	Weightage
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Assignments	20%
Lab Exercises	20%
Quizzes	10%
Mini Project	20%
Capstone Project	30%

Learning Outcomes

By the end of the course, learners will be able to:

- Understand the fundamentals of C++ programming and object-oriented design.
- Develop efficient programs using functions, arrays, pointers, and dynamic memory allocation.
- Design reusable software using classes, objects, inheritance, polymorphism, and abstraction.
- Handle exceptions and perform file operations for data persistence.
- Utilize the Standard Template Library (STL) for efficient data management and algorithms.
- Implement basic data structures and solve real-world programming problems.
- Debug, test, and optimize C++ applications using industry-standard development tools.
- Build modular, scalable, and maintainable software applications using modern C++ programming practices.